CMVP Submission Checklist — pqc_crypto_module v0.1.0

> **Disclaimer**: Prepared for FIPS 140-3 evaluation, not currently validated.

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1. Required CMVP Artifacts

#	Artifact	Status	Owner	Location
1	Security Policy	READY	[TBD]	`crates/pqc_crypto_module/SECURITY_POLICY.md`
2	Design Document	READY	[TBD]	`crates/pqc_crypto_module/DESIGN_DOCUMENT.md`
3	Finite State Model	READY	[TBD]	`crates/pqc_crypto_module/FINITE_STATE_MODEL.md`
4	Key Management	READY	[TBD]	`crates/pqc_crypto_module/KEY_MANAGEMENT.md`
5	Self-Test Documentation	READY	[TBD]	`crates/pqc_crypto_module/SELF_TEST_DOCUMENTATION.md`
6	Non-Approved Usage	READY	[TBD]	`crates/pqc_crypto_module/NON_APPROVED_USAGE.md`
7	Boundary Definition	READY	[TBD]	`crates/pqc_crypto_module/build/module_boundary_definition.md`
8	Reproducible Build	READY	[TBD]	`crates/pqc_crypto_module/build/reproducible_build.md`
9	Test Coverage Report	READY	[TBD]	1500+ tests across 12 suites; `cargo test`
10	Operational Guidance	READY	[TBD]	`crates/pqc_crypto_module/OPERATIONAL_GUIDANCE.md`
11	Configuration Management Plan	READY	[TBD]	`crates/pqc_crypto_module/CONFIGURATION_MANAGEMENT.md`

2. Pending Artifacts

#	Artifact	Status	Owner	Notes
11	NIST Official Test Vectors	NEEDS WORK	[TBD]	Internal KATs exist; NIST ACVP vectors not yet integrated
12	CAVP Algorithm Certificates	NEEDS WORK	[TBD]	Requires lab engagement; ML-DSA, SHA3-256, ML-KEM
13	Lab Engagement	NOT STARTED	[TBD]	No lab selected; see `LAB_SELECTION.md`
14	Entropy Source Documentation (SP 800-90B)	NEEDS WORK	[TBD]	OsRng used; compliance documentation pending
15	Vendor Evidence Package	NOT STARTED	[TBD]	Lab-specific format; prepared after lab selection

3. Pre-Submission Verification

- [x] All approved algorithms documented with standards references (FIPS 202/203/204)
- [x] Non-approved algorithms identified, gated, and documented
- [x] State machine: Uninitialized -> SelfTesting -> Approved -> Error
- [x] Boundary definition: 11 source files in `src/`, Rust crate isolation
- [x] Self-tests: KATs for ML-DSA-65, SHA3-256 at startup
- [x] Key zeroization: `ZeroizeOnDrop` on all key types
- [x] Approved-mode enforcement: `require_approved()` guard on all operations
- [x] `approved-only` Cargo feature excludes legacy module at compile time
- [ ] Official NIST ACVP test vectors integrated
- [ ] CAVP certificates obtained for each approved algorithm
- [ ] Lab selected and engagement initiated
- [ ]

SP 800-90B entropy source analysis completed

4. Document Review Status

Document	Internal Review	External Review
Security Policy	Complete	Pending lab
Design Document	Complete	Pending lab
Finite State Model	Complete	Pending lab
Key Management	Complete	Pending lab
Self-Test Documentation	Complete	Pending lab
Non-Approved Usage	Complete	Pending lab
Boundary Definition	Complete	Pending lab
Reproducible Build	Complete	Pending lab

5. Next Steps

1. Integrate NIST ACVP official test vectors (see `TEST_VECTOR_PLAN.md`)
2. Select accredited lab (see `LAB_SELECTION.md`)
3. Prepare contact package for lab outreach (see `CONTACT_PACKAGE/`)
4. Address gap analysis findings (see `GAP_ANALYSIS.md`)
5. Begin CAVP algorithm certificate process with selected lab